Lecture Notes:
Select Topics in Living Systems

Based on lectures by **Dr. Ram Adar** in 2025–26
Notes created by Daniel Haim Breger

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These notes were typed during the 2026 winter semester *Select Topics in Living Systems* course taught by **Dr. Ram Adar** at the Technion.

The lectures were given in Hebrew and were live-translated by me to English. The document is provided as is and likely contains many errors.

If you find any mistakes or typos please let me know at `danielbreger@campus.technion.ac.il`.

Contents

Lecture 1.

Wed, October 29, 2025

1 Phase Transitions in Equilibrium

1.1 Motivation

Phase is a distinct state of matter which can be described by an "order parameter". We'll focus on isotropic polar phase transition. In this case the order parameter is the polarization $\mathbf{p} = \langle \hat{n} \rangle$ where $\hat{n}$ is the unit vector of the particles' direction.

Phase transition

In certain areas of physics we see systems which aren't ordered at high temperatures but become ordered at low temperatures. Why? A competition between energy and entropy (Free energy being $F = U - TS$). The temperature at which order emerges is called the critical temperature, marked T_C and the change in the system is called a phase transition.

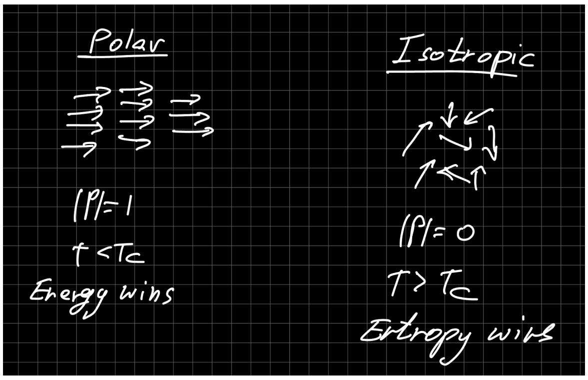

Figure 1.1: Polar and Isotropic ordering of matter.

1.2 Physical Description

We'll assume free energy of the following form:

$$F(T, N, \psi) = F_0(T, N) + F_\psi(T, N, \psi) \tag{1.1}$$

where ψ is the order parameter. ψ is not a conserved quantity but rather a degree of freedom of the system (in contrast to the number of particles, for example). Therefore under equilibruim we find

$$F_{eq}(T, N, \psi) = F_0(T, N) + F_\psi(T, N, \psi_{eq}) \tag{1.2}$$

and the equilibrium value is determined by

$$\left.\frac{\partial F_\psi}{\partial \psi}\right|_{\psi=\psi_{eq}} = 0 \tag{1.3}$$

and the following is true:

$$\psi_{eq}(T > T_C) = 0 \tag{1.4}$$

$$\psi_{eq}(T < T_C) \neq 0 \tag{1.5}$$

We'll focus on phase transition of 2nd order, for which the order parameter changes continuously, thus $\psi_{eq}(T = T_C) = 0$. Otherwise the phase transition is of first order.

1.3 Landau Theory of Phase Transitions

The theory was developed by Lev Landau, for which he earned the Nobel Prize in Physics in 1962. It states that close to the critical temperature T_C the physics of phase transitions is universal. For a phase transition of 2nd order, we will write F_ψ in a phenomenological form as a Taylor series expansion:

$$F_\psi = \frac{a}{2}\psi^2 + \frac{b}{4}\psi^4 \tag{1.6}$$

The terms being of even powers because we don't care about directions. The quadratic term's coefficient is of the form

$$a = a_0 \cdot \frac{T - T_C}{T_C} \tag{1.7}$$

Where $a_0 > 0$. This quantifies the competition between energy (which wins at $T < T_C$) and entropy (which wins at $T > T_C$). The parameter $b > 0$ ensures stability: ψ_{eq} receives finite values and does not approach infinity. Because of a's sign, we find $\psi_{eq} \neq 0$ for $T < T_C$. Explicitly:

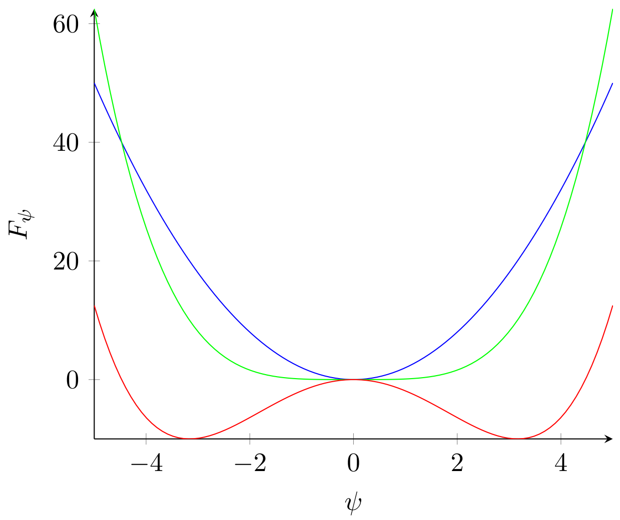

Figure 1.2: The graphs for the functions $2x^2$ (blue), $0.1x^4$ (green) and $0.1x^4 - 2x^2$ (red).

$$0 = \left.\frac{\partial F}{\partial \psi}\right|_{\psi=\psi_{eq}} = a\psi_{eq} + b\psi_{eq}^3 = b\psi_{eq}\left(\psi_{eq}^2 + \frac{a}{b}\right) \tag{1.8}$$

therefore

$$\psi_{eq} = 0, \pm\sqrt{\frac{a_o}{b} \cdot \frac{T - T_C}{T_C}} \tag{1.9}$$

There is no distinction between the positive and negative sign. Either can be chosen arbitrarily (spontaneous symmetry break). There will be a preference given an external field which will add an element of the form $-h\psi$ to the free energy.

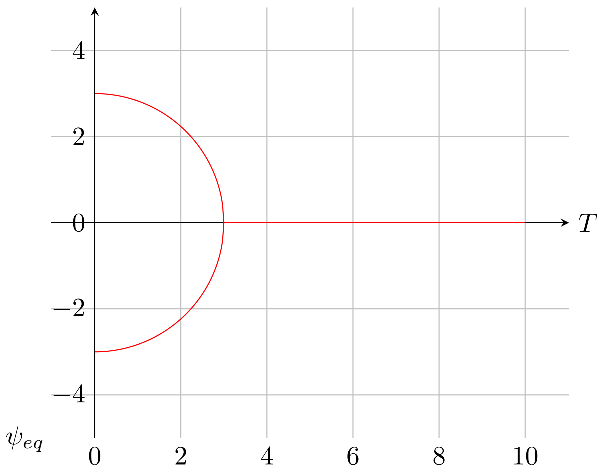

Figure 1.3: The possible solutions from equation 1.9.

Where the parameter a comes from

We'll assume a simple model of polarisation with energy

$$U = -\sum_{i,j \quad neighbors} u n_\alpha^i n_\alpha^j, \quad u > 0 \tag{1.10}$$

The energy is lower when i,j point in the same direction. Using mean field approximation the internal energy is

$$U = \frac{u}{2} \sum_{i,j} \langle n_\alpha^i \rangle \cdot \langle n_\alpha^j \rangle = -\frac{u}{2} N Z p_\alpha^2 \tag{1.11}$$

where $p_\alpha = \langle n_\alpha \rangle$, N is the number of particles, and Z the number of close neighbors. As for the entropy, we can find from a taylor expansion

$$S = S(p = 0) - \alpha N p^2 \tag{1.12}$$

and summing it up we find

$$F = U - TS = N \left(\alpha T - \frac{u}{2} Z \right) p^2 = a_0 \cdot \frac{T - T_C}{T_C} p^2 \tag{1.13}$$

where

$$T_C = \frac{uZ}{2\alpha} \qquad\qquad a_0 = \alpha N T_C \tag{1.14}$$

If instead of the vector n_α there is spin $(\pm)$ then we find a similar result in the mean field framework. This model is called the Ising model, one of the most basic and useful models in statistical mechanics.

A dynamical description of the phase transition

We described the phase transition from the free energy but we can also describe it from dynamics. For a non-conserved order parameter we can write the dynamics as follows:

$$\dot{\psi} = -\Gamma_\psi \frac{\partial F}{\partial \psi}, \quad \Gamma_\psi > 0 \tag{1.15}$$

the idea being that the dynamics guarantee that the free energy only gets smaller (the total entropy grows). We can see this using the chain rule:

$$\dot{F} = \frac{\partial F}{\partial \psi} \dot{\psi} = -\Gamma_\psi \left(\frac{\partial F}{\partial \psi} \right)^2 < 0 \tag{1.16}$$

this dynamics is called Model A (and often appears with a "noise" parameter). We'll insert F and find:

$$\dot{p} = -\Gamma_p(ap + bp^3) \tag{1.17}$$

the fixed points are $\frac{\partial F}{\partial p} = 0$, the extremums of the free energy. The stability of the solution is determined in phase space. in the language of dynamical systems, a phase transition is called a

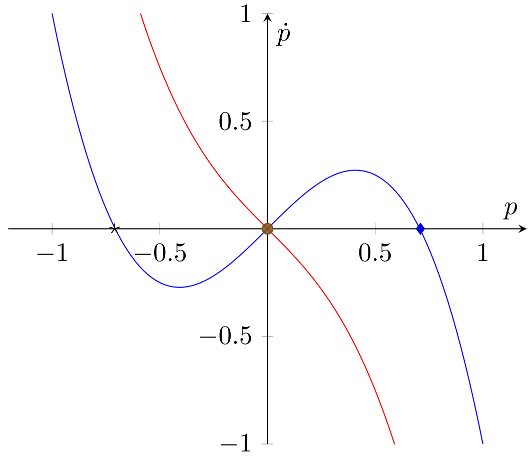

Figure 1.4: The graphs of $f(x) = x - 2x^3$ (blue) and $g(x) = -x - 2x^3$ (red). The positive and negative points marked on the x axis are stable points of equilibrium for $f(x)$. The point marked at (0,0) is a stable point for $g(x)$ but an unstable one for $f(x)$

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bifurcation. This approach is useful for out of equilibrium systems, for which we can't write down a free energy. Later on in the course, we'll come across dynamical equations of a similar form. In general dynamical systems, when a function F whose derivative is strictly non-positive with a finite minimum exists (like free energy), then it's called a Lyapunov function.

1.4 Spatial Dependance: The Ginsburg-Landau Theory

We saw that the direction of $\mathbf{p}$ is arbitrary, in practice different areas of the system will initially develop order in different directions. For simplicity, we'll assume $\mathbf{p} = p\hat{y}$ where p can be either positive or negative. We'll look at the case where $\mathbf{p}$ points down at $x \to -\infty$ and up at $x \to \infty$.

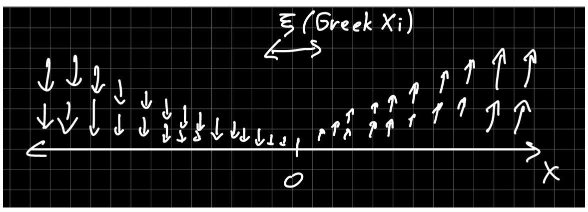


Figure 1.5: A smooth phase transition with Domain Wall ξ.

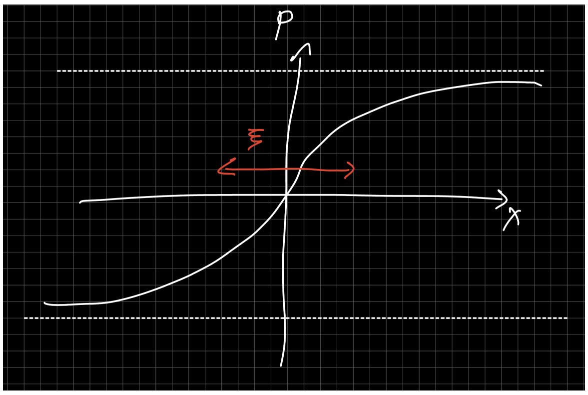


Figure 1.6: A smooth phase transition with Domain Wall ξ.

We'll find a finite range around 0 of width χ in which $|\mathbf{p}| \neq p_{eq}$. This region is called the Domain Wall. To quantify the cost to change $\mathbf{p}$ in space we'll write the density of free energy per length unit:

$$F[p] = \int \mathrm{d}x f = \int \mathrm{d}x \left(\frac{a}{2}p^2 + \frac{b}{4}p^4 + \frac{k}{2}p'^2 \right) \tag{1.18}$$

where $k > 0$. The last element quantifies the cost to change p. In higher dimensions we'll have $(\nabla p)^2$. Why is it like this? The cost comes from the gradient, and given there is no preferred direction, a positive and negative change should cost the same.

Lecture 2.

Wed, November 5, 2025

The profile $p(x)$ is at the point where the functional derivative (with deltas) of F with respect to p is 0, i.e

$$0 = \frac{\partial f}{\partial p} - \left(\frac{\partial f}{\partial p'}\right)' \; ; F = \int \mathrm{d}x f \tag{2.1}$$

If we apply this to our specific functional we find

$$0 = \frac{\delta F}{\delta p} = ap + bp^3 - kp'' \tag{2.2}$$

which if we simplify it we find

$$\frac{k}{a}p'' = p + \frac{b}{a}p^3 \tag{2.3}$$

if we focus on the case where $a < 0$ and $p_{eq}^2 = \frac{L}{a}$ then

$$-\frac{k}{|a|}p'' = p\left(1 - \frac{p^2}{p_{eq}^2}\right) \tag{2.4}$$

which using dimensional analysis we found a length scale that goes with $\sqrt{\frac{k}{|a|}}$. Looking back at figure 1.6 we guess a solution of the form

$$p_0(x) = p_{eq}\tanh\frac{x}{\xi} \tag{2.5}$$

and use the following identity

$$\tanh' x = \frac{1}{\cosh^2 x} = 1 - \tanh^2 x \tag{2.6}$$

we find

$$p'' = \frac{p_{eq}}{\xi}\left(1 - \tanh^2\frac{x}{\xi}\right)' \tag{2.7}$$

$$= -2\frac{p_{eq}}{\xi^2}\tanh\frac{x}{\xi}\left(1 - \tanh^2\frac{x}{\xi}\right) \tag{2.8}$$

$$= -\frac{2}{\xi^2}p_0\left(1 - \frac{p_0^2}{p_{eq}^2}\right) \tag{2.9}$$

and from this we conclude

$$\frac{2}{\xi^2}\frac{k}{|a|} = 1 \to \xi = \sqrt{\frac{2k}{|a|}} \tag{2.10}$$

in particular, near the phase transition ($a \approx 0$), ξ diverges. Additionally, the excess free energy in the ξ width layer in comparison to the energy in phases at $\pm p_{eq}$ is called the interfacial tension (γ). It can be shown (in homework) that

$$\gamma = \int kp_0'^2\,\mathrm{d}x = \sqrt{\frac{-8ka^3}{9b^2}} \tag{2.11}$$

1.5 The Mermin-Wagner (-Hohenberg) Theorem

So far we dealt with spatial change of a binary order parameter p, like spin, such that $\mathbf{p} = p\hat{y}$ but in a more general case $\mathbf{p} = p(\cos\theta, \sin\theta)$ and θ can change continuously. We'll notice that

$$F_p = \frac{a}{2}|\mathbf{p}||^2 + \frac{b}{4}|\mathbf{p}||^4 \tag{2.12}$$

is invariant under a change of θ. This is a continuous symmetry and θ is called the Goldstone/soft mode.

The Theorem Itself

A system under equilibrium at temperature $T > 0$ of dimension $d \leq 2$ will NOT develop a long range order which breaks a continuous symmetry.

Explanation

Let's look at how the free energy is dependent on θ. There is no homogeneous contribution but rather only from derivatives. As before, we'll write

$$F = \frac{k}{2}\int \mathrm{d}\mathbf{r}(\nabla\theta)^2 \tag{2.13}$$

Looking at a configuration such as

How much free energy does it cost to create a continuous change in θ in a system of dimension d with characteristic length L? From dimensional considerations $\nabla^2 \sim L^{-2}$ and $\int \mathrm{d}\mathbf{r} \sim\sim L^d$ therefore $F \sim L^{d-2}$. For a large L and $d \leq 2$ the cost is low. Therefore change in θ will happen in a spontaneous manner from thermal fluctuations at $T > 0$.

A couple notes

1. In this course we will focus on *active systems out of equilibrium.* The theorem is violated and we can achieve long range order in 2D. This was the first result in active matter which created significant interest in the field.

2. Even in equilibrium a phase transition under the theorem's conditions is possible ($d = 2$, $T > 0$). In this case the order is not exactly long-range and is characterized by topological defects of the order parameter (we'll deal with defects later in the course). Sometimes such a transition is viewed as a KT transition (Kosterlitz-Thouless, 2016 Nobel prize in Physics)

2 Matter Out of Thermal Equilibrium

2.1 Types of Systems Out of Thermal Equilibrium

In general two main types of systems out of thermal equilibrium exist:

Non-Equilibrium Steady State In this case an outside energy source is injecting energy into the system, mostly through the system's edge, which dissipates. Examples:

1. Shear flow: Energy flows through the externally driven surfaces and dissipates through viscosity.
2. Electric circuits: A source is driving an electric current through the circuit, which dissipates at the resistor.

Near-Equilibrium Systems The system was previously in thermal equilibrium, which it left due to a controlled change in a thermodynamical parameter (T,P) or of an external potential. We can characterize the dynamics of the system arriving at the new equilibrium (relaxation). Examples:

1. Moving an optical trap: a colloid is captive in an optical trap which is suddenly moved. How does the colloid's position change over time?
2. Polymers stretching: a polymer is held on both ends by a force f. The force is suddenly changed. How will the edge-to-edge length of the polymer change?

In this course we will focus on active matter in which the energy is consumed locally by the system's elements. We will mostly focus on the case of a steady state. In general, given a system with measurable dynamics (such as a biological system), we will want to know whether it is in thermal equilibrium or not.

2.2 Macroscopic Description of Movement

In thermal equilibrium a closed system can only carry out a combination of uniform linear motion and rigid body rotation. Therefore, movement out of equilibrium will mostly be characterized by:

Wet Active Matter A closed system with non-uniform motion, such as a liquid with shearing. A relevant active system example for this case is active particles in a liquid.

Dry Active Matter An open system with uniform or non-uniform motion. A relevant active system example for this case is active particles on a surface.

Intuition

A system in which heat is created is out of equilibrium, since its entropy grows. In the first example heat is created due to viscosity and in the second due to friction. In an active system heat can be created from the active mechanism itself too (e.g a chemical reaction).

2.3 Proof of the statement in 2.2

We'll section a body into small but macroscopic parts with mass, energy, and momentum m_i, E_i, p_i respectively. The entropy results from the internal energy, and not from the kinetic energy of the center of mass.

$$S = \sum_i S_i(E_i - \frac{p_i^2}{2m_i^2}) \tag{2.14}$$

In a closed system the angular and linear momenta are conserved:

$$\sum_i \mathbf{p_i} = const. \qquad\qquad \sum_i \mathbf{r_i} \times \mathbf{p_i} = const. \tag{2.15}$$

In thermal equilibrium the entropy is maximized while conforming to the constraints.

$$0 = \frac{\partial}{\partial \mathbf{p_i}} \sum_j \left[S_j \left(E_j - \frac{p_j^2}{2m_j} \right) + \mathbf{a} \cdot \mathbf{p}_j + \mathbf{b} \cdot (\mathbf{r}_i \times \mathbf{p}_j) \right] \tag{2.16}$$

$$= -\frac{1}{T} \frac{\mathbf{p}_i}{m_i} + \mathbf{a} + \mathbf{b} \times \mathbf{r}_i \tag{2.17}$$

$$\rightarrow \mathbf{v}_i = \mathbf{u} + \boldsymbol{\Omega} \times \mathbf{r}_i \tag{2.18}$$

where

$$\mathbf{u} = T \cdot \mathbf{a} \qquad \qquad \boldsymbol{\Omega} = T \cdot \mathbf{b} \tag{2.19}$$

2.4 Detailed Balance & Time Reversal Symmetry

Detailed Balance

Assume a system with microstates $\{ s \}$ and a probability function P_s which changes with time. Its dynamics are given by a Master Equation:

$$\frac{\partial}{\partial t} P_s = \sum_{s'} (P_{s'} W_{s' \to s} - P_s W_{s \to s'}) \tag{2.20}$$

where $W_{s \to s'}$ is the transition rate between s and s'. At equilibrium a detailed balance exists, each element in the sum cancels out:

$$P_{s'} W_{s' \to s} = P_s W_{s \to s'} \tag{2.21}$$

Sometimes this is instead written as

$$\frac{P_s}{P_{s'}} = \frac{W_{s' \to s}}{W_{s \to s'}} \tag{2.22}$$

Lecture 3.

Wed, November 19, 2025

Example: Movement on a 1D lattice

Let's look at a 1-dimensional lattice with N sites such that the probability to be at each site is $1/N$. For a random walk (such as diffusion)

$$W_{i,i+1} = W_{i+1,i} \tag{3.1}$$

since the probability of moving in either direction is equal. This results in a detailed balance. Next let's assume an active particle with a "positive spin" (i.e it moves to the right), therefore

$$W_{i,i+1} > W_{i+1,i} \tag{3.2}$$

and as a result there is no detailed balance. It is possible to connect the rate $W_{i,i+1}$ compared to $W_{i+1,i}$ to the difference between a step forward or backwards in time. Therefore systems with time reversal symmetry also are in detailed balance.

Time Reversal Symmetry

Systems in Equilibrium are symmetric to time reversal. If we watch a video of their dynamics and run it backwards, we won't feel a difference. This isn't necessarily true for active particles (like swimming fish). We'll assume a system with a hamiltonian which is a function of the momenta p^N (e.g $H = \sum\limits_i \frac{p^2}{2m} + U(\mathbf{r})$) and that there is no magnetic field or global rotation which will add linear elements to the momenta. The dynamics of this system are given by Hamilton's equations:

$$\frac{\mathrm{d}r_k}{\mathrm{d}t} = \frac{\partial H}{\partial p_k} \qquad\qquad \frac{\mathrm{d}p_k}{\mathrm{d}t} = -\frac{\partial H}{\partial r_k} \tag{3.3}$$

Since H is an even function w.r.t p, $\frac{\partial H}{\partial p_k}$ is odd. Looking at

$$t = -\bar{t} \qquad\qquad r_k = \overline{r_k} \qquad\qquad p_k = -\overline{p_k} \tag{3.4}$$

we see that we'll get the same equations for $\bar{t}, \overline{r_k}, \overline{p_k}$. What does this mean? Assume we start moving in phase space at the point (r^N, p^N). After any time period τ we'll reach the point $(r^{N'}, p^{N'})$. Next we'll start at the point $(r^{N'}, p^{N'})$, and after another time period τ we'll reach (r^N, p^N). This is called time reversal symmetry. When there is no time reversal symmetry the system is out of equilibrium. For example, looking at an active particle in 2D with velocity

$$\frac{\mathrm{d}\mathbf{r}_k}{\mathrm{d}t} = v(\cos\theta_k, \sin\theta_k) \tag{3.5}$$

where θ is a random variable distributed between 0 and 2π. When we reverse time, the left side of the equation will inverse but the right side won't.

2.5 The Fluctuation Dissipation Theorem

In thermal equilibrium or near it there is a fundamental connection between thermal fluctuations and the reaction to external forces, and it's called the FDT (Fluctuation-Dissipation Theorem). It has many versions and many implications and it is one of the most important and useful results in statistical mechanics. The connection between spontaneous fluctuations and reactions to external disturbances was identified by Lars Onsager, and he received the 1968 Nobel prize in chemistry for it (Onsager's regression hypothesis)

Static Version (In equilibrium)

We'll assume a system with hamiltonian H_0 and we'll apply an external force f with a tied field x such that $H = H_0 - fx$. In equilibrium:

$$\langle x \rangle = \frac{\sum_i x_i e^{-\beta(H_0 - f x_i)}}{e^{-\beta(H_0 - f x_i)}} \tag{3.6}$$

therefore

$$\frac{\partial \langle x \rangle}{\partial \beta f} = \frac{\sum_i x_i^2 e^{-\beta H}}{\sum_i e^{-\beta H}} - \langle x \rangle \frac{\sum_i e^{-\beta H} x_i}{\sum_i e^{-\beta H_i}} \tag{3.7}$$

noticing the first element is the average of x_i squared we get

$$k_B T \frac{\partial \langle x \rangle}{\partial f} = \langle x^2 \rangle - \langle x \rangle^2 \tag{3.8}$$

Near Equilibrium

We'll assume a system near equilibrium such that the deviation from equilibrium is linear in the field which takes it out of equilibrium. We'll assume a quantity x with a time independent equilibrium average $\langle x \rangle$. Still, x performs thermal fluctuations such that $\delta x(t) = x(t) - \langle x \rangle$ while $\langle \delta x \rangle = 0$. We can learn information about the system from its time correlation function:

$$C(t, t') = \langle \delta x(t) \delta x(t') \rangle = [\text{in equilibrium}] = C(t - t') \tag{3.9}$$

therefore we can write

$$C(t) = \langle \delta x(0) \delta x(t) \rangle = \langle x(0) x(t) \rangle - \langle x \rangle^2 \tag{3.10}$$

which is a generalization of the variance to include time dependence. The averages used above are for different starting conditions

$$C(t) = \int \mathrm{d}r^N \, \mathrm{d}p^N \delta x(0) \delta x(t) g(r^N, p^N) \tag{3.11}$$

in thermal equilibrium g is defined by the Boltzmann distribution. Looking at a system in which a field f is acting starting at $t' \to -\infty$ such that at $t = 0$ the system is in thermal equilibrium with $H = H_0 - fx$. We'll assume that at time $t = 0$ the field is switched off. We find that

$$k_B T \frac{\overline{x}(t) - \langle x \rangle}{f} = \langle x(t) x(0) \rangle - \langle x \rangle^2 \tag{3.12}$$

where $\overline{x}(t)$ is the average given by the starting conditions out of equilibrium. A full development can be found in the appendixes.

In Terms of Reaction Function

In general

$$\delta\overline{x}(t) = \int_{-\infty}^{\infty} \mathrm{d}t' \chi(t,t') f(t') \tag{3.13}$$

where χ is a reaction function. χ is determined by the properties of the system in equilibrium (taking into consideration corrections of order f to δx) therefore $\chi(t,t') = \chi(t-t')$, there is no meaning to absolute time but only for the time since the disturbance. In addition, the disturbance only affects the future (causality). We'll define

$$f(t) = \begin{cases} f & t < 0 \\ 0 & t \geq 0 \end{cases} \tag{3.14}$$

and we found that

$$\langle \delta x(0) \delta x(t) \rangle = \frac{k_B T}{f} \overline{\delta x} = \frac{k_B T}{f} \int_{-\infty}^{0} \mathrm{d}t' \chi(t-t') f \tag{3.15}$$

$$= k_B T \int_{t}^{\infty} \mathrm{d}t'' \chi(t'') \tag{3.16}$$

differentiating we find

$$\frac{\mathrm{d}C}{\mathrm{d}t} = -k_B T \chi(t), \quad t \geq 0 \tag{3.17}$$

Which is a more popular formulation of the same. We'll look at yet another alternative formulation. In an experiment we can examine whether FDT happens (if the system is near thermal equilibrium) by looking at the spectrum of the response and correlation functions.

$$\hat{C}(\omega) = \int_{-\infty}^{\infty} \mathrm{d}t C(t) e^{i\omega t} \qquad \hat{\chi}(\omega) = \int_{0}^{\infty} \mathrm{d}t \chi(t) e^{i\omega t} = \chi'(\omega) + i\chi''(\omega) \tag{3.18}$$

and we find

$$\hat{C}(\omega) = \frac{2k_B T}{\omega} \hat{\chi}''(\omega) \tag{3.19}$$

Why is $\hat{\chi}''$ dissipation? For a periodic force $f = f_0 \cos(\omega t)$ we find

$$\chi(t) = Re[\chi(\omega) f(\omega) e^{-i\omega t}] \tag{3.20}$$

$$= \hat{\chi}'(\omega) f_0 \cos \omega t + \hat{\chi}''(\omega) f_0 \sin \omega t \tag{3.21}$$

$$\rightarrow \dot{x}(t) = -f_0 \hat{\chi}'(\omega) \cdot \omega \sin \omega t + f_0 \hat{\chi}' \cdot \omega \cos \omega t \tag{3.22}$$

then the average dissipation in a period T is

$$\frac{1}{T} \int_{0}^{T} \mathrm{d}t f(x) \dot{x}(t) = \frac{1}{2} f_0^2 \omega \hat{\chi}''(\omega) \tag{3.23}$$

Lecture 4.

Wed, November 26th, 2025

2.6 Example: Brownian Motion & The Einstein Relation

Assume a system of particles, specifically colloids (large brownian particle) which is moved with an external force F in 1 dimension. The particle's motion is given by

$$\dot{x} = \mu F + v_B \tag{4.1}$$

where μ is the colloid's mobility, v_B is the noise element of brownian motion. This noise element has the following properties

$$\langle v_B(t)\rangle = 0 \qquad \langle v_B(t)v_B(t')\rangle = 2D\delta(t-t') \tag{4.2}$$

where D is the diffusion parameter. Without a force we will find

$$\langle x(t)\rangle = \int_0^t \mathrm{d}t' \langle v_B(t')\rangle = 0 \tag{4.3}$$

as the expectation value of a sum is the sum of the expectation values. Additionally,

$$\langle x^2\rangle = \left\langle \int_0^t \mathrm{d}t' v_B(t') \int_0^t \mathrm{d}t'' v_B(t'') \right\rangle \tag{4.4}$$

$$= \int_0^t \mathrm{d}t' 2D \int_0^t \mathrm{d}t'' \delta(t'-t'') = 2Dt \tag{4.5}$$

which is the same result we would find from other methods in considering a random walk. To find the response function we can notice that

$$-w\omega\tilde{x} = \mu\tilde{F} \tag{4.6}$$

therefore

$$\tilde{\chi}(\omega) = \frac{i\mu}{\omega} \implies \tilde{\chi}'' = \frac{\mu}{\omega} \tag{4.7}$$

If we compare this to the correlation function

$$C_v(t) = \langle v_B(t)v_B(0)\rangle = 2D\delta(t) \implies \tilde{C}_v = 2D \tag{4.8}$$

and

$$v = \dot{x} \implies \tilde{v} = i\omega\tilde{x} \implies \tilde{C}_v(\omega) = \omega^2\tilde{C}_x(\omega) \tag{4.9}$$

and therefore

$$\boxed{\tilde{C}_x(\omega) = \frac{2D}{\omega^2}} \tag{4.10}$$

and now applying the FDT

$$\frac{2D}{\omega^2} = \frac{2k_BT}{\omega}\hat{\chi}''(\omega) = 2k_BT\frac{\mu}{\omega^2} \tag{4.11}$$

implies

$$\boxed{D = k_BT\mu} \tag{4.12}$$

This concludes the introduction prior to discussing active matter.

3 Hydrodynamic Theory Of Polar Active Matter

3.1 Hydrodynamic Theory And Slow Variables

The hydrodynamic theory is a continuum theory made to describe the changes in fields over large time scales and large distances. Such a theory can be developed in two main ways:

Start From a Microscopic Model and perform coarse-graining.

Directly Write Macroscopic equations based on conservation laws and symmetries.

We will focus on the 2nd approach. Later in the course we will present a standardized method to develop such a theory from considerations of entropy production.

Slow Variables

We'll assume a set of fast variables $\{ \psi_k \}$ and slow variables $\{ \varphi_i \}$ which satisfy equations of the form

$$\frac{\partial \psi_k}{\partial t} = \frac{1}{\tau_f} G_k(\{\psi_j\}, \{\varphi_j\}) \tag{4.13}$$

$$\frac{\partial \varphi_i}{\partial t} = \frac{1}{\tau_s} F_i(\{\psi_j\}, \{\varphi_j\}) \tag{4.14}$$

where τ_s and τ_f are time constants which satisfy

$$\tau_f = O(1) \qquad\qquad \tau_s = O(L^\alpha) \tag{4.15}$$

with $\alpha > 0$ and L the size of the system. For large systems we'll find separation on time scales. For large systems we'll write

$$G_k(\{\psi_j\}, \{\varphi_j\}) = 0 \tag{4.16}$$

and we'll express the fast variables in terms of the slow variables. Explicitly, we'll normalize times using τ_s such that

$$\tilde{t} = \frac{t}{\tau_s} \tag{4.17}$$

and we'll find

$$\frac{\partial \psi_k}{\partial \tilde{t}} = \frac{\tau_s}{\tau_f} G_k(\dots) \tag{4.18}$$

$$\frac{\partial \varphi_i}{\partial t} = F_i(\dots) \tag{4.19}$$

and since

$$\frac{\tau_s}{\tau_f} = O(L^\alpha) >> 1 \tag{4.20}$$

we'll find that to leading order

$$G_k = 0 \tag{4.21}$$

next we express $\{ \psi_k \}$ in terms of $\{ \varphi_i \}$ and find

$$\frac{\partial \varphi_i}{\partial t} = \frac{1}{\tau_s} \overline{F}(\{\varphi_j\}) + \xi_i(t) \tag{4.22}$$

where

$$\overline{F}(\{\varphi_j\}) = F(\{\varphi_j\}, \{\psi_k(\{\varphi_j\})\}) \tag{4.23}$$

The influence of the fast variables comes into effect through the noise function ξ_i which has $\langle \xi_i \rangle = 0$. For short correlation times it is customary to assume

$$\langle \xi_i(t) \xi_j(t') \rangle \propto \delta_{ij} \delta(t - t') \tag{4.24}$$

in this course we'll focus on an average description without the noise. Otherwise the theory is called Fluctuating Hydrodynamics.

Types Of Slow Variables

Conserved Quantities They satisfy a continuum equation

$$\frac{\partial \varphi}{\partial t} = -\nabla \cdot \jmath \tag{4.25}$$

without a spatial change in φ there is no current therefore

$$\jmath = -\alpha \nabla \varphi \tag{4.26}$$

we can insert this and find

$$\frac{\partial \varphi}{\partial t} = \alpha \nabla^2 \varphi + \dots \tag{4.27}$$

we'll perform a fourier transform

$$\tilde{\varphi}_{\mathbf{q}} = \int \mathrm{d}\mathbf{r} e^{-i\mathbf{q}\cdot\mathbf{r}} \varphi(\mathbf{r}) \tag{4.28}$$

therefore

$$\frac{\partial \tilde{\varphi}_{\mathbf{q}}}{\partial t} = -\alpha q^2 \tilde{\varphi}_{\mathbf{q}} + \dots \tag{4.29}$$

in conclusion, the time to decay to mode $\mathbf{q}$ is

$$\tau_{\mathbf{q}} \sim q^{-2} \tag{4.30}$$

The hydrodynamic theory focuses on large length scales (small q-s). The smallest q is $\frac{\pi}{L}$, therefore $\tau \sim L^2$. In a more general case, the dynamics of a conserved quantity is described by Model B

$$\frac{\partial \varphi}{\partial t} = -\nabla \cdot \jmath \tag{4.31}$$

$$\jmath = -\gamma \nabla \frac{\delta F}{\delta t} + \xi \tag{4.32}$$

$$F[\varphi] = \int \mathrm{d}\mathbf{r} f(\varphi, \nabla \varphi) \tag{4.33}$$

where F is the free energy functional, γ is a kinetic parameter, and **xi** is noise which satisifies

$$\langle \xi_i \rangle = 0 \tag{4.34}$$

and

$$\langle \xi_i(t) \xi_j(t') \rangle = 2 k_B T \gamma \delta_{ij} \delta(t - t') \tag{4.35}$$

and we'll notice that

$$\dot{F} = \int \frac{\delta F}{\delta \varphi} \dot{\varphi} \, \mathrm{d}r = \gamma \int \frac{\delta F}{\delta \varphi} \partial_\alpha \partial_\alpha \frac{\delta F}{\delta \varphi} \, \mathrm{d}\mathbf{r} \tag{4.36}$$

$$= -\gamma \int \left(\partial_\alpha \frac{\delta F}{\delta \varphi} \right) \left(\partial_\alpha \frac{\delta F}{\delta \varphi} \right) < 0, \qquad \gamma > 0 \tag{4.37}$$

in a classical hydrodynamic theory (NS) the conserved quantities are the mass and momentum.

Soft/Goldstone Mode When a free energy is invariant w.r.t a continuous quantity, that quantity is called a soft mode. We mentioned this when we discussed the MW theorem when we looked at the mode x (magnetization with continuous angle). We'll look at two configurations which have a different structure but the same free energy. both have the same free energy, therefore there is no reason for one to decay into the other. The typical time to decay will be proportional to the size of the system. The dynamics of such quantities which aren't conserved are described by Model A

$$\frac{\partial \varphi}{\partial t} = -\gamma \frac{\delta F}{\delta \varphi} + \xi \tag{4.38}$$

where

$$\dot{F} = \int \frac{\delta F}{\delta \varphi} \frac{\partial \varphi}{\partial t} \, \mathrm{d}r = -\gamma \int \left(\frac{\delta F}{\delta \varphi} \right)^2 \mathrm{d}r < 0, \qquad \gamma > 0 \tag{4.39}$$

so

$$F = \int \mathrm{d}\mathbf{r} \left(\frac{1}{2} K \left(\nabla \varphi \right)^2 + \dots \right) \tag{4.40}$$

where

$$\frac{\delta F}{\delta \varphi} = -k \nabla^2 \varphi \tag{4.41}$$

and again we find a diffusion equation with $\tau \sim L^2$.

Lecture 5.

Wed, December 3rd, 2025

Order Parameter Near Critical Points (Type 2 Transition)

If we look at the Ginzburg-Landau free energy

$$F = \int d\mathbf{r} \left(\frac{a}{2}\varphi^2 + \frac{b}{4}\varphi^4 + \frac{K}{2} (\nabla\varphi)^2 \right) \tag{5.1}$$

near the critical point $(a = 0)$ $\varphi = 0$ and the free energy almost does not change with φ, similarly to the soft mode. The slowing down of the dynamics near a critical point is called *critical slowing down.*

4 Hydrodynamic Theory For Polar Active Matter On a Surface (Toner Tu, 1995)

As a reminder, a polarization field is

$$\mathbf{P} = \langle \mathbf{n} \rangle \tag{5.2}$$

what are the slow parameters?

Mass / Number of Particles are a conserved quantity

Momentum? NO! Since we're interacting with a surface there is friction, thus momentum is not conserved.

The direction $\frac{\mathbf{P}}{|\mathbf{P}|}$ is a soft mode.

$|\mathbf{P}|$ is a slow parameter near the critical points if P changes continuously in phase transition.

We'll write equations for ρ and $\mathbf{P}$ as if they were passive and add phenomenology of active elements. Active matter on a surface is called "dry". Later in the course we'll deal with "wet" active matter too in which the particles are in a liquid medium and there is conservation of momentum.

4.1 Equations for Density

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot \jmath \qquad \jmath = -\gamma_\rho \nabla \frac{\delta F}{\delta \rho} + \rho v_0 \mathbf{P} \quad \gamma_\rho > 0 \tag{5.3}$$

where $\frac{\delta F}{\delta \rho}$ is the chemical potential and $\rho v_0 \mathbf{P}$ is the active current element which is equivalent to assuming speed $v\hat{n}$ for each particle in the absence of gradients. How did we define this element? We need to find a vector, and the only vectors in the system are $\mathbf{P}$ and ∇. In the hydrodynamic limit $\nabla \sim L^{-1}$ and is small in comparison to $\mathbf{P}$. In addition, many gradient element can be found from the passive element.

4.2 Equation for Polarization

$$\frac{\partial \mathbf{P}}{\partial t} + \lambda_1 (\mathbf{P} \cdot \nabla) \mathbf{P} = -\gamma_P \frac{\delta F}{\delta \mathbf{P}} \tag{5.4}$$

where $\lambda_1 (\mathbf{P} \cdot \nabla) \mathbf{P}$ is an element added which is similar to the advection element in Navier-Stokes $(\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{u}(t, \mathbf{x}) = \partial_t \mathbf{u} + (\mathbf{v} \cdot \nabla) \mathbf{u})$. In an active system $\mathbf{P}$ plays a double role of both an order parameter and of the velocity such that its equation is similar to NS. Why didn't we choose other elements? Elements without derivatives can be found from $\frac{\delta F}{\delta \mathbf{P}}$. Other elements with first order derivatives we can also find from F. λ_1 is a phenomenological element and doesn't necessarily satisfy $\lambda_1 = v_0$ like in NS. This is because there is no conservation of momentum or Galileo invariance.

4.3 Isotropic-Polar Transition (Flocking Transition)

If we look at a homogenous system, the equation for **P** is equivalent to a passive system. The isotropic-polar transition will be determined by the form of the free energy. We'll write down free energy like in Landau theory where the density is the critical parameter:

$$F = \int d\mathbf{r} \left(\frac{a}{2} \left(1 - \frac{\rho}{\rho_c} \right) P^2 + \frac{b}{4} P^4 + \dots \right) \quad b > 0 \tag{5.5}$$

for $\rho < \rho_c$ a minimum F at $P = 0$ and the system is isotropic.

for $\rho \geq \rho_c$ we'll find a minimum $P_0 = \sqrt{\frac{-a}{b}}$ and then the system is polar.

In a more general manner the transition can be described depending on the density and the effective temperature T. In hydrodynamic theory, F will include a contribution from an ordering interaction like in the XY model, in which case $1/T$ acts as the strength of the interaction. In the Vicsek model, T is tied to the strength of the noise in determining the direction of the particles.

4.4 What's Special About the Active System?

Coexistence has special dynamics of traveling bands. Additionally, the stability of the polar phase is different to that of a passive system. In particular, in contradiction of the MW theorem there can be long range order which breaks continuous symmetry in 2D. How is this possible? First, this is a system out of equilibrium, so the theorem doesn't apply. In as simple terms as possible, we'll explain how the polar phase is stable thanks to the active advection element $(\lambda_1 (\mathbf{P} \cdot \nabla) \mathbf{P})$.

We'll analyze the instability in a passive system in a new way. We'll assume an ordered system with $\theta = 0$ and a point deviation at $\mathbf{r} = 0$ to the angle $\delta\theta$. Since it's a soft mode, the deviation decays diffusively $(\partial_t \delta\theta \sim \nabla^2 \delta\theta)$. In d dimensions at time τ, the deviation is "smeared" over the volume $V_e \sim \tau^{d/2}$. Deviations are constantly created, so at time τ there are

$$n_e \sim V_e \cdot \tau \sim \tau^{1+d/2} \tag{5.6}$$

deviations in the volume. The deviations are in random directions. From the central limit theorem, the standard deviation satisfies

$$\Omega_e \sim \sqrt{n_e} \sim \sqrt{\tau V_e} \tag{5.7}$$

and the average deviation per particle is

$$\Delta\theta \sim \frac{\Omega_e}{V_e} \sim \sqrt{\frac{\tau}{V_e}} \sim r^{1-d/2} \tag{5.8}$$

where $\tau \sim \sqrt{r}$, and in the thermodynamic limit r is large. We found $\Delta\theta$ is insignificant for $d > 2$, and grows with the system's size for $d < 2$. The case $d = 2$ is marginal and gives $\Delta\theta \sim l_n L$. This is similar to the calculation at equilibrium. It is important that the order is continuous because noise causes small fluctuations, in contrast to domain wall energy.

Lecture 6.

Wed, December 10th, 2025

In the active case, the volume V_{int} is determined not only by diffusion but also by advection. A deviation $\delta\theta$ creates different separation between particles in the $\hat{n} \parallel \hat{p}$ direction and the $\hat{n} \perp \hat{p}$ direction. This gives us

$$\hat{P} = (\cos\theta, \sin\theta) \approx \left(1 - \frac{1}{2}\delta\theta^2, \delta\theta\right) \tag{6.1}$$

and

$$\delta x_\perp \sim v_0 \tau \sin \delta\theta \sim \tau\delta\theta \qquad\qquad \delta x_\parallel \sim v_0 \tau \delta\theta^2 \sim \tau\delta\theta^2 \tag{6.2}$$

and the volume

$$V_e \sim \omega_\perp^{d-1} \omega \parallel \tag{6.3}$$

From a combination of advection and diffusion:

$$\omega_\perp \sim \delta x_\perp + D_\perp \tau^{1/2} \sim \tau\delta\theta + D_\perp \tau^{1/2} \sim \tau^{\gamma_\perp} \tag{6.4}$$

and

$$\omega_\parallel \sim \delta x_\parallel + D_\parallel \tau^{1/2} \sim \tau\delta\theta^2 + D_\parallel \tau^{1/2} \sim \tau^{\gamma} \tag{6.5}$$

if we define $\delta\theta \sim \tau^\gamma$ then we also find

$$\delta\theta \sim \sqrt{\frac{\tau}{V_e}} \sim \frac{\tau^{1/2}}{\sqrt{\omega_\perp^{d-1}\omega \parallel}} \sim \tau^\gamma \tag{6.6}$$

We thus have to solve three equations for the gammas:

$$2\gamma = 1 - \gamma_\parallel - (d-1)\,\gamma_\perp \tag{6.7}$$

$$\gamma_\perp = \max\left(1+\gamma, \frac{1}{2}\right) \tag{6.8}$$

$$\gamma_\parallel = \max\left(1+2\gamma, \frac{1}{2}\right) \tag{6.9}$$

we find $\gamma < 0$ for all $d > 1$. Specifically for $d = 2$ we find

$$\gamma = -\frac{1}{5} \qquad\qquad \gamma_\perp = \frac{4}{5} \qquad\qquad \gamma_\parallel = \frac{3}{5} \tag{6.10}$$

This means the order is stable, thanks to the 'quick smearing' in the direction perpendicular to $\mathbf{P}$. A couple of notes:

- The stability of polar phase is still being researched in a more rigorous manner.
- Current research shows that active polar order has additional properties. For example, a boundary effect can affect order in bulk (Yariv Kafri).

4.5 Waves in Active Polar Matter

Remembering the equations

$$\frac{\partial P}{\partial t} = -\partial_\alpha j_\alpha \qquad j_\alpha = -\gamma_\rho \partial_\alpha \frac{\delta F}{\delta \rho} + v_0 \rho P_\alpha \qquad \frac{\partial P_\alpha}{\partial t} + \lambda\left(P_\beta \partial_\beta\right) = -\gamma_P \frac{\delta F}{\delta P_\alpha} \tag{6.11}$$

We'll examine small deviations around the homogenous stable state

$$\rho = \rho_0 + \delta\rho(t, \mathbf{r}) \qquad\qquad \mathbf{P} = \mathbf{P_0} + \delta\mathbf{P}(t, \mathbf{r}) \tag{6.12}$$

for

$$\frac{\delta\rho}{\rho_0}, \frac{|\mathbf{P}|}{|\mathbf{P}|} \ll 1 \tag{6.13}$$

we'll linearize the equations and keep leading elements in the derivatives (first order). For simplicity we'll assume $\mathbf{P} = \hat{x} + \delta P \hat{y} \implies$ the steady state is completely ordered and only rotations of the polarizations are possible. In this case, there is a steady current $\jmath_{\mathbf{0}} = \rho_0 v_0 \hat{x}$. In this case we won't interest ourselves with the Landau elements of the free energy ($\frac{a}{2}P^2 + \frac{b}{4}P^4$).

$$F = \int \mathrm{d}^2 r \left[\frac{\chi}{2} \left(\frac{\delta\rho}{\rho_0} \right)^2 + \frac{1}{2} k \partial_\alpha P_\beta \partial_\alpha P_\beta + \left(\omega_0 - \omega_1 \frac{\delta\rho}{\rho_0} \right) \partial_\alpha P_\alpha \right] \tag{6.14}$$

What does the last element mean? From integration by parts

$$F = \int \mathrm{d}^2 r \left[\cdots - P_\alpha \partial_\alpha \left(\omega_0 - \omega_1 \frac{\delta\rho}{\rho_0} \right) \right] + \text{boundary element} \tag{6.15}$$

$$= \int \mathrm{d}^2 r \left[\cdots + \frac{\omega_1}{\rho} \mathbf{P} \cdot \nabla \rho \right] \tag{6.16}$$

ω_0 only contributes a boundary element. ω_1 determines whether $\mathbf{P}$ likes to order itself with or against the gradients in the density. We'll develop the equation in 6.11 to first order in $\delta\rho, \delta P$ and up to derivatives of first order and find

$$\partial_t \delta\rho = -\partial_\alpha \left(v_0 \delta\rho P_\alpha^0 + v_0 \rho_0 \delta P_\alpha \right) \tag{6.17}$$

where

$$\left(\partial_t + v_0 \partial_x \right) \delta\rho \approx -v_0 \rho_0 \partial_y \delta P \tag{6.18}$$

$$\left(\partial_t + \lambda \partial_x \right) \delta P \approx -\gamma_P \omega_1 \partial_y \frac{\delta\rho}{\rho_0} \tag{6.19}$$

because

$$\frac{\delta F}{\delta P_\alpha} = -k \partial_{\beta\beta} \delta P_\alpha + \frac{\omega_1}{\rho_0} \partial_\alpha \delta\rho \tag{6.20}$$

If the system is homogenous in the y direction, we find that $\delta P, \delta\rho$ are independent up to first order. One moves at velocity v_0 and the other at velocity λ. If the system is homogenous in the x direction, we find

$$\partial_t \delta\rho = -v_0 \rho_0 \partial_y \delta P \tag{6.21}$$

$$\partial_t \delta P = -\gamma_P \omega_1 \partial_y \frac{\delta\rho}{\rho_0} \tag{6.22}$$

which gives us

$$\partial_{tt} \delta\rho = v_0 \gamma_P \omega_1 \partial_{yy} \delta\rho \tag{6.23}$$

for $\omega_1 > 0$ we found a wave equation with the velocity

$$c = \sqrt{\omega_1 \gamma_P v_0} \tag{6.24}$$

Changes in the density reorganize $\mathbf{P}$ and changes in $\mathbf{P}$ transport the density because of the activity. Note: the higher order terms in derivatives will lead to decay of the waves, i.e to diffusion.

5 Nematic Active Matter

5.1 The Nematic Order Parameter

So far we dealt with polar order $\mathbf{P} = \langle \hat{l} \rangle$ with a distinct direction. Sometimes there is only a distinct axis. This kind of order is called nematic. The order parameter is the Nematic Tensor:

$$Q_{\alpha\beta} = \left\langle \hat{l_\alpha} \hat{l_\beta} - \frac{\delta_{\alpha\beta}}{d} \right\rangle = Q \left(n_\alpha n_\beta - \frac{\delta_{\alpha\beta}}{d} \right) \tag{6.25}$$

Its properties are

1. Invariant to the substitution $\hat{n} \iff -\hat{n}$
2. It is symmetric and traceless

For $d = 2$ and complete order along the $\hat{x}$ axis:

$$Q_{\alpha\beta} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} \tag{6.26}$$

in general, in the direction $\theta \to \hat{n} = (\cos\theta, \sin\theta)$

$$Q_{\alpha\beta} = \frac{1}{2} \begin{pmatrix} \cos(2\theta) & \sin(2\theta) \\ \sin(2\theta) & -\cos(2\theta) \end{pmatrix} \tag{6.27}$$

with the angle being 2θ representing the fact that $\hat{n} \iff -\hat{n}$ ($\theta \iff \theta + \pi$). A physical theory of nematic matter can be also written in terms of $\hat{n}$ while maintaining the symmetry for $\hat{n} \iff -\hat{n}$ and a scalar parameter Q. In practice, it is customary to sometimes use $\mathbf{P}$ instead of $\hat{n}$. Remember this does not mean there is a preferred direction. Even if we can write $\left\langle \hat{l}_\alpha \right\rangle$ we can write

$$Q_{\alpha\beta} = P_\alpha P_\beta - P_\mu P_\mu \frac{\delta_{\alpha\beta}}{d} \tag{6.28}$$

and in nematic theory the symmetry $\mathbf{P} \iff -\mathbf{P}$ must be maintained. What does that mean? Let's return to the equations.

$$F = \int \mathrm{d}\mathbf{r} \left[\frac{a}{2} P^2 + \frac{b}{4} P^4 + \frac{K}{2} \partial_\alpha P_\beta \partial_\alpha P_\beta - \cancel{\omega(\rho) \partial_\alpha P_\alpha} + \dots \right] \tag{6.29}$$

The final element does not fit the symmetry we require so we cannot use it in nematic theory.

$$\frac{\partial \rho}{\partial t} = -\partial_\alpha j_\alpha \qquad j_\alpha = -\gamma_\rho \partial_\alpha \frac{\delta F}{\delta \rho} + \cancel{v_0 \rho P_\alpha} \tag{6.30}$$

the final element in the right hand equation also cannot be used for the same reason. Finally

$$\frac{\partial P_\alpha}{\partial t} + \cancel{\lambda (P_\mu \partial_y) P_\alpha} = -\gamma_P \rho \frac{\delta F}{\delta P_\alpha} \tag{6.31}$$

The unintuitively cancelled element is $(\mathbf{P} \cdot \nabla) \mathbf{P}$. Why is it unallowed? Intuitively it comes from advection which no longer exists. Mathematically, we'll multiply by P_β and find

$$P_\beta \partial_t P_\alpha + \lambda P_\beta (P_\mu \partial_\mu) P_\alpha = \dots \tag{6.32}$$

$$\partial_t Q_{\alpha\beta} + \cancel{\lambda (P_\mu \partial_\mu) Q_{\alpha\beta}} = \dots \tag{6.33}$$

What is the active contribution? We need to take into consideration active currents. How can we build it from $Q_{\alpha\beta}$? We need to apply a vector on it. Without a defined direction, the only vector is ∇:

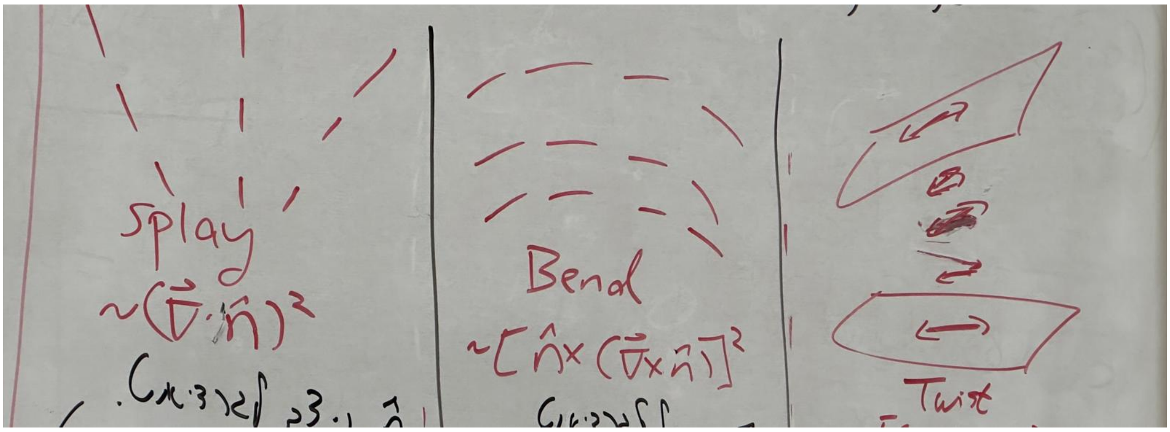
$$\jmath = -\gamma_\rho \nabla \frac{\delta F}{\delta \rho} - \Lambda \nabla \cdot \mathbf{Q} \tag{6.34}$$

Lecture 7.

Wed, December 24th, 2025

5.2 Deformations in Nematic Order: Free Energy

As we've seen so far, it's useful to understand the free energy it takes to deform the order parameter. We'll imagine a completely ordered state, such that changes in the nematic order are only the result of changes in the angle. It is possible to develop the elements in the free energy phenomenologically from symmetry. In the nematic case we find three elements.



Splays The elements have the form $\sim (\nabla \cdot \hat{n})^2$. $\hat{n}$ is perpendicular to the gradient and $\delta\hat{n}$ is parallel to the gradient.

Bends The elements have the form $(\hat{n} \times (\nabla \times \hat{n}))^2$. $\hat{n}$ is parallel to the gradient and $\delta\hat{n}$ is perpendicular ot the gradient.

Twists This is only defined in three dimensions. The elements have the form $(\hat{n} \cdot (\nabla \times \hat{n}))^2$. Both $\hat{n}$ and $\delta\hat{n}$ are perpendicular to the gradient.

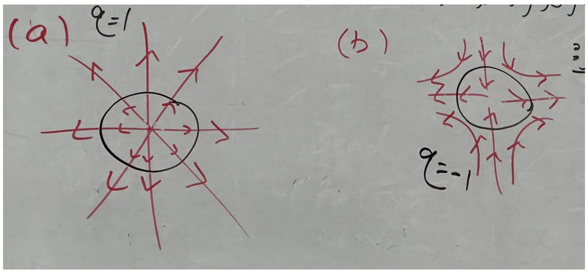
in each of the cases the element $\hat{n}$ appears an even number of times and the derivatives are of second order. In total, the deformation energy we find is called the Frank Free Energy or the Frank Elasticity:

$$F = \int \mathrm{d}\mathbf{r} \frac{1}{2} \left(K_1 (\nabla \cdot \hat{n})^2 + K_2 (\hat{n} \cdot (\nabla \times \hat{n}))^2 + K_3 (\hat{n} \times (\nabla \times \hat{n}))^2 \right) \tag{7.1}$$

the coefficients $K_{1,2,3}$ are called the Frank coefficients and in general they have different values but are of similar order of magnitude. For simplicity it is possible to assume $K_1 = K_2 = K_3$. This is called the 'one-constant approximation' in which case it is then possible to express the free energy as

$$F = \int \mathrm{d}\mathbf{r} \frac{K}{2} \partial_\alpha n_\beta \partial_\alpha n_\beta = \int \mathrm{d}\mathbf{r} \frac{K}{2} (\nabla \hat{n})^2 \tag{7.2}$$

in practice the final expression has another boundary element called the sadlle-splay. We'll return to the active current $-\Lambda \partial_\beta Q_{\alpha\beta}$ in 2D. We'll look at how a current can result from a splay or a bend.



It is possible to think of each line as a force dipole. When they are non-homogenous there are regions which will contribute to the creation of a total force (like electric dipoles and charge). For case 1 in the above figure:

$$\hat{n} = \hat{r} \implies \partial_\beta (n_\alpha n_\beta) = n_\alpha \partial_\beta n_\beta + \cancel{n_\beta \partial_\beta n_\alpha} = 2\hat{r} \tag{7.3}$$

and for case 2:

$$\hat{n} = \hat{\theta} \implies \partial_\beta (n_\alpha n_\beta) = \cancel{n_\alpha \partial_\beta n_\beta} + n_\beta \partial_\beta n_\alpha = \frac{1}{r}\partial_\theta (-\sin\theta, \cos\theta) = -\frac{\hat{r}}{r} \tag{7.4}$$

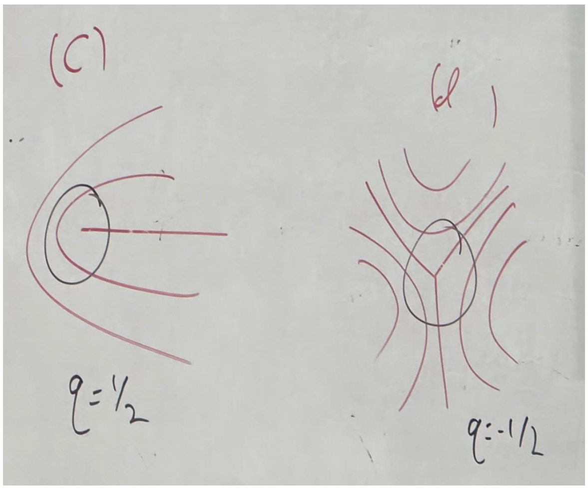
6 Topological Defects (in 2D)

Unlike the deformations we saw already, there are deformations with singularities in the direction $\hat{r}$, i.e places where the direction is undefined. Such deformations are called topolotical defects. We'll focus on them in 2 dimensions. It is possible to characterize defects by the 'topological charge' q. It is called as the winding number. We'll assume that $\hat{n} = (\cos\theta, \sin\theta)$. In that case the topological charge is given by

$$\boxed{\oint \mathrm{d}\theta = 2\pi q} \tag{7.5}$$

where the integral is over any closed path which encloses the defect and its direction is counter-clockwise. For polar order, q is a whole number. A couple notes:

- The value of q is independent of the path.
- If there are two or more defects enclosed by the path, q will be the total charge.



6.1 The Free Energy of a Defect

We'll employ the Frank free energy with the one-constant approximation

$$F = \frac{1}{2}\int \mathrm{d}^2 r K (\nabla n)^2 = \frac{1}{2}\int \mathrm{d}\mathbf{r} K (\nabla\theta)^2 \tag{7.6}$$

this is analogous to electrostatics where the angle plays the rule of the electric potential and the defect will be like a charge. The configuration of θ around the defect results from a minimization of the free energy:

$$0 = \frac{\delta F}{\delta\theta} = -K\nabla^2\theta \tag{7.7}$$

This is a local minimum. The global minimum is a constant zero. We'll solve the laplace equation we found in polar coordinates:

$$0 = \frac{1}{r}\partial_r (r\partial_r\theta) + \frac{1}{r^2}\partial_{\varphi\varphi}\theta \tag{7.8}$$

Assuming that $\theta = \theta(\varphi)$

$$\theta = \theta_0 + a\varphi \tag{7.9}$$

We'll enforce the charge:

$$2\pi q = \oint \mathrm{d}\theta = \int_0^{2\pi} \mathrm{d}\varphi \frac{\mathrm{d}\theta}{\mathrm{d}\varphi} = 2\pi a \tag{7.10}$$

i.e

$$\boxed{\theta = \theta_0 + q\varphi} \tag{7.11}$$

Now let's calculate the free energy:

$$\nabla\theta = \frac{q}{r}\hat{\varphi} \implies F = \int_0^\infty \mathrm{d}r r \int_0^{2\pi} \frac{1}{2} K \frac{q^2}{r^2} \mathrm{d}\varphi \tag{7.12}$$

$$= \pi K q^2 \int_0^\infty \frac{\mathrm{d}r}{r} \implies F = \pi \, K q^2 \ln r \Big|_0^\infty \tag{7.13}$$

this diverges at both limits! At the limit going to infinity, this is a manufactured divergence since the system will have a finite size R. At the limit of $r = 0$, our theory isn't applicable to short distances (hydrodynamic continuum theorey). We also assumed the order parameter is of magnitude 1 with only the direction changing, because of the energetic cost to change the magnitude. But close to $r = 0$, changing the direction costs so much energy that the order parameter will prefer to shrink. The solution to this problem is that we'll define a 'core' of radius a such that there's an F_{core} which we can't calculate exactly, and we'll take a as the lower limit of the integral. Therefore

$$\boxed{F = F_{core} + \pi K q^2 \ln \frac{R}{a}} \tag{7.14}$$

like in electrostatics, the energy goes with q^2. This means that we'll often find defects with the smallest q possible in equilibrium (± 1 for polar, $\pm 1/2$ for nematic). Moreover, defects with opposite charges will attract and will eventually merge to minimize the free energy. The result $F \sim \ln \frac{R}{a}$ is very important in the theory of condensed matter. The entropy of the location of defects also goes $\sim \ln \frac{R}{a}$. This gives us $F \sim \left(\frac{\pi K}{4} - k_B T\right) \ln \frac{K}{q}$. At sufficiently high temperatures defects will begin to form. This is the basis for topological phase transitions (KT/KTHN). The topology of the system has influence on the total charge of the defects (The poincare-Hopf theorem) in particular, a nematic field on the surface of a sphere must have a total topological charge $q = 2$.

6.2 Dynamics of Active Nematic Defects

We defined in active nematic matter a current

$$j^a_\alpha = -\Lambda \partial_\beta Q_{\alpha\beta} \tag{7.15}$$

we'll examine this element for defects $q = \pm 1/2$. Intuition: from a simple look at the defects, $q = 1/2$ seems to have a directionality, and $q = -1/2$ doesn't, so only the first will move.

Lecture 8.

Wed, December 31st, 2025

The nematic tensor is

$$Q = \begin{pmatrix} \cos(2\theta) & \sin(2\theta) \\ \sin(2\theta) & -\cos(2\theta) \end{pmatrix} \tag{8.1}$$

a proof was given as homework. Therefore the current for the nematic tensor is

$$\begin{aligned} \mathbf{j^a} &= -\Lambda\left(\partial_x Q_{xx} + \partial_y Q_{xy}, \partial_x Q_{yx} + \partial_y Q_{yy}\right) && (8.2)\\ &= -\Lambda\left(-\sin(2\theta)\partial_x\theta + \cos(2\theta)\partial_y\theta, \cos(2\theta)\partial_x\theta + \sin(2\theta)\partial_y\theta\right) && (8.3) \end{aligned}$$

theta's derivatives are, given that $\theta = q\varphi$

$$\begin{aligned} \partial_x\theta &= q\partial_x \tan^{-1}\frac{y}{x} = q\frac{1}{1+\left(\frac{y}{x}\right)^2} - \frac{y}{x^2} = -q\frac{y}{x^2+y^2} && (8.4)\\ &= -\frac{q}{r}\sin\varphi && (8.5) \end{aligned}$$

and

$$\partial_y\theta = \cdots = \frac{q}{r}\cos\varphi \tag{8.6}$$

inserting back into the current

$$\begin{aligned} \mathbf{j^a} &= -\Lambda\frac{q}{r}\left(\sin(2q\varphi)\sin\varphi + \cos(2q\varphi)\cos\varphi, -\cos(2q\varphi)\sin\varphi + \sin(2q\varphi)\cos\varphi\right) && (8.7)\\ &= -\Lambda\frac{q}{r}\left(\cos\left((2q-1)\varphi\right), \sin\left((2q-1)\varphi\right)\right) && (8.8) \end{aligned}$$

if we insert $q = \pm 1/2$

$$\mathbf{j^a} = \frac{\Lambda}{2r}\begin{cases} -\hat{x} & q = 1/2 \\ \cos(2\varphi)\hat{x} - \sin(2\varphi)\hat{y} & q = -1/2 \end{cases} \tag{8.9}$$

if we average over the edge of the core, $r = a$

$$\frac{1}{2\pi}\int_0^{2\pi} \mathrm{d}\varphi \mathbf{j^a}(r = a, \varphi) = \begin{cases} \frac{-\Lambda}{2a}\hat{x} & q = 1/2 \\ 0 & q = -1/2 \end{cases} \tag{8.10}$$

where $\hat{x}$ was chosen as the axis of symmetry ($\theta(\varphi = 0) = 0$). Several conclusion:

1. A $-1/2$ defect doesn't move.
2. A $1/2$ defect moves along the axis of symmetry. For a positive Λ, which is called extensile matter, it moves with the direction of the head. For contractile matter for which Λ is negative, it moves in the direction of the tail.

This expresses whether the particles create a repulsive or attractive force dipole. Later we'll show that $-\Lambda Q_{\alpha\beta}$ is the active stress acting in the system.

7 The Stress Tensor and Forces in Liquids

7.1 Motivation and Definition of the Stress Tensor

So far we dealt with active matter on a surface, in which case momentum is exchanged between the particles and surface, like in walking. Since the surface isn't described, momentum is not conserved.

Such systems are called 'Dry Active Matter'. In general, in a liquid medium momentum is transferred between the active matter and the liquid, and is overall conserved. In this case the active particles are called 'Swimmers' and the systems belong to 'Wet Active Matter'. When momentum is conserved, it is possible to write a continuity equation:

$$\boxed{\partial_t (\rho v_\alpha) = \partial_\beta \sigma^{tot}_{\alpha\beta}} \tag{8.11}$$

the tensor in the right-hand side is the total stress tensor. We'll look at an integral over a small volume V:

$$\int_V \partial_t (\rho v_\alpha) \, \mathrm{d}V = \int_V \partial_\beta \sigma^{tot}_{\alpha\beta} \, \mathrm{d}V \tag{8.12}$$

$$= \oint_{\partial V} \sigma^{tot}_{\alpha\beta} n_\beta \, \mathrm{d}S \tag{8.13}$$

it is possible to conclude that $\sigma^{tot}_{\alpha\beta}$ is the force in the α direction per unit surface in the direction β which acts on the volume element. Since the stress is the negative of the momentum flux, it contains an element of the form $-\rho v_\alpha v_\beta$. This element is called the Reynolds element. We'll break up the stress into an isotropic element and a traceless element

$$\boxed{\sigma^{tot}_{\alpha\beta} = -P\delta_{\alpha\beta} + \tilde{\sigma}_{\alpha\beta} - \rho v_\alpha v_\beta} \tag{8.14}$$

7.2 Viscous Liquid vs Elastic Solid

A solid has a defined shape. It deforms in response to external force, and will return to its original form when the force is removed. A liquid has no defined shape and will change its shape without limit. We'll see what this has to do with the stress using the following experiment:

A material is placed between two surfaces at a distance of h between them. The bottom surface is held in place and the top is connected using a wire to a weight which applies force F at time $0 \leq t \leq t_0$ and at time t_0 the wire is cut. At time $0 \leq t \leq t_0$ a stress $\tilde{\sigma} = \frac{F}{S}$ acts where S is the surface of the material. This is shear stress. Due to the stress, strain is formed in the material which can be defined as $\gamma = \frac{d}{h}$.

In an ideal elastic solid (Hookean, like a spring) the stress is linear in the straing: $\tilde{\sigma} = G\gamma$, and G is called the shear modulus. In an ideal viscous liquid (also called Newtonian Liquid) the stress is connected to the rate of strain: $\tilde{\sigma} = \eta\dot{\gamma}$ where η is called the viscousity. In a liquid, shearing is an irreversible process which creates heat. The branch that describes how materials flow and deform is called Rheology. Soft materials sometimes are not pure solids or pure liquids and are called 'visco-elastic'. The simplest model for viscousity is the maxwell model in which an ideal spring ($\tilde{\sigma} = G\gamma$) is connected in series to an ideal liquid ($\tilde{\sigma} = \eta\dot{\gamma}$). The strain contains a contribution from both the spring and liquid, and the stress on each element is equal.

$$\dot{\gamma} = \dot{\gamma_1} + \dot{\gamma_2} = \frac{\dot{\sigma}}{G} + \frac{\sigma}{\eta} \implies (1 + \tau\partial_t)\,\sigma = \eta\dot{\gamma} \tag{8.15}$$

where $\tau = \frac{\eta}{G}$ is the typical decay time for stress. The solution is

$$\sigma(t) = G \int_0^t \mathrm{d}t' \dot{\gamma} e^{-\frac{t-t'}{2}} \tag{8.16}$$

in short time periods we get behavior similar to that of a solid in long times to that of a liquid. The difference between different materials is prominent in the case of fast strain.

7.3 Rheology of Living Systems

The rest of the course will focus on liquid rather than solids, because cells/tissue act as a liquid over long time periods. Why? There are 2 processes which do not permit a 'solid state of reference' and cause currents over time: cell division and T1 processes.

Cell Division When cells divide they push each other and cause a current. At time scales larger than a typical timeframe (24 hours), cells act as q liquid.

T1 processes When cells cover a surface in 2D (confluent monolayer) they take on a typical shape of a hexagon, simiilarly to foam. This can be understood from the minimization of the surface tension between the cells in a given area. Still, due to division, initial disorder, and other processes the cells are dynamic. T1 processes are processes of neighbor exchanges which allow flow and decay of elastic strains which require the deformation of cells.

7.4 The Stress in Liquids and the Stokes Equation

We saw that in liquids the shear stress is $\sigma = \eta\dot{\gamma}$. In general the stress depends on the velocity v_α. Since there are no relative friction forces between different parts of the liquid which flows at uniform velocty, we need derivatives of the velocity. In hydrodynamic theory we deal with large length scales, so we'll assume the stress depends only on the first derivatives of the velocity, and that it's linear in them. Additionally, since there are no relative forces in uniform rotation ($\mathbf{v} = \mathbf{\Omega} \times \mathbf{r}$) we'll find that the stress depends only on the symmetric part of $\partial_\alpha v_\beta$:

$$v_{\alpha\beta} = \frac{1}{2}\left(\partial_\alpha v_\beta + \partial_\beta v_\alpha\right) \tag{8.17}$$

this tensor is sometimes called the strain rate.

Lecture 9.

Wed, January 7th, 2026

In isotropic liquids:

$$\sigma^{tot}_{\alpha\beta} = -P\delta_{\alpha\beta} - \rho v_\alpha v_\beta + 2\eta v_{\alpha\beta} \tag{9.1}$$

where the first element is a pressure element, the second is advection, and the third is viscousity. If we insert this into the momentum conservation equation we get

$$\partial_t (\rho v_\alpha) = \partial_\beta (-P\delta_{\alpha\beta} - \rho v_\alpha v_\beta + 2\eta v_{\alpha\beta}) \tag{9.2}$$

which is kind of like the naviet stokes equation. We'll simplify this. We'll compare the 2 last elements and assume a length scale L:

$$\eta v_{\alpha\beta} \sim \eta \frac{v}{L} \tag{9.3}$$

$$\rho v_\alpha v_\beta \sim r v^2 \tag{9.4}$$

the ratio between these is called the Reynolds number

$$Re = \frac{\rho v^3}{\eta \frac{v}{L}} = \frac{vL}{\nu} \tag{9.5}$$

where $\nu = \frac{\eta}{\rho}$ is caled the kinematic viscousity. When $Re \ll 1$, the dynamics is controlled by the viscousity rather than advection. We'll estimate Reynolds in a system of cells. We'll take density to be that of water $\rho \approx 10^3\,\mathrm{kg/m^3}$ and $\eta = 10^3\,\mathrm{Pa}$, which gives us $\nu = 10^{19}$ something which gives us a very small Reynolds number. We'll thus focus on $Re \ll 1$ and ignore $\rho v_\alpha v_\beta$. We'll focus on stationary cases where $\partial_t (\rho v_\alpha) = 0$. We get an equation of forces called the Stokes equation

$$0 = f_\alpha = \partial_\beta (-P\delta_{\alpha\beta} + 2\eta v_{\alpha\beta}) \tag{9.6}$$

Many liquids like water are incompressible. In that case (in which ρ is constant in space)

$$0 = \partial_t \rho = -\partial_\alpha (\rho v_\alpha) = -\rho \partial_\alpha v_\alpha \implies \partial_\alpha v_\alpha = 0 \tag{9.7}$$

In this case P acts as a lagrange multiplier, guarnteeing incompressibility.

8 Wet Active Matter And Entropy Production

We'll want to write down a hydrodynamic theory for wet active matter. The first question is what the slow quantities. Our conserved quantities are:

$$\partial_t \rho = -\partial_\alpha (\rho v_\alpha) \tag{9.8}$$

$$\partial_t (\rho v_\alpha) = \partial_\beta \sigma^{tot}_{\alpha\beta} \tag{9.9}$$

and we are in a soft mode, so the angle of the polar/nematic order. (We'll look at ordered system with $|\mathbf{P}| = 1$).

8.1 Entropy Production

We'll look at a system in thermal contact with a large temperature reservoir R at equilibrium at temperature T. The system exchanges energy with the reservoir while consering the total energy.

$$\mathrm{d}U_R + \mathrm{d}U = 0 \tag{9.10}$$

and the change in total entropy is

$$\mathrm{d}S_T = \mathrm{d}S + \mathrm{d}S_R \tag{9.11}$$

since the reservoir is in thermal equilibrium,

$$\mathrm{d}S_R = \frac{\mathrm{d}U_R}{T} = -\frac{dU}{T} \tag{9.12}$$

so the total entropy

$$\mathrm{d}S_T = \mathrm{d}S - \frac{\mathrm{d}U}{T} = -\frac{1}{T}\left(T\,\mathrm{d}S - \mathrm{d}U\right) = -\frac{1}{T}\,\mathrm{d}F \tag{9.13}$$

so the entropy production rate is

$$\frac{\mathrm{d}}{\mathrm{d}t}S_T = -\frac{1}{T}\frac{\mathrm{d}}{\mathrm{d}t}F \geq 0 \tag{9.14}$$

the inequality being the 2nd law of thermodynamics. How will we see the production of entropy in the systems we are interested in? We'll first assume a passive homogenous system with no average currents, of volume V, density ρ, and order parameter $\mathbf{p}$.

$$F = F(\rho, \mathbf{p})\ ;\ \mathrm{d}F = V\left(\mu\,\mathrm{d}\rho - \mathbf{h}\cdot\mathrm{d}\mathbf{p}\right) \tag{9.15}$$

where $\mu = \frac{\delta F}{\delta \rho}$ is the chemical potential and $\mathbf{h} = -\frac{\delta F}{\delta \mathbf{p}}$ is melocular/ordering field. From the chain rule, the change in time is

$$-T\frac{\mathrm{d}S_T}{\mathrm{d}t} = \frac{\mathrm{d}F}{\mathrm{d}t} = \int \mathrm{d}\mathbf{r}\left(\mu\partial_t\rho - h_\alpha\partial_t p_\alpha\right) \tag{9.16}$$

Now we'll take into consideration a system with a current and we'll find

$$T\dot{S} = \int \mathrm{d}\mathbf{r}\left(\sigma_{\alpha\beta}v_{\alpha\beta} + h_\alpha\frac{D}{D_t}p_\alpha\right) \tag{9.17}$$

where

$$\sigma_{\alpha\beta} = \sigma_{\alpha\beta}^{tot,sym} + \rho v_\alpha v_\beta + P\delta\alpha\beta \tag{9.18}$$

$$\frac{D}{Dt}P_\alpha = \left(\partial_t + v_\beta\partial_\beta\right)P_\alpha + \omega_{\alpha\beta}P_\beta \tag{9.19}$$

$$h_\alpha = -\frac{\delta F}{\delta P_\alpha} \tag{9.20}$$

the first line is deviatoric stress, the 2nd line is called the co-rotational derivative, and the 3rd is the molecular field. What about activity? In a biological system the main source of activity is hydrolysis of ATP. We'll denote the energy produced in the process as $\Delta\mu \approx 20k_BT$ and we'll denote r as the rate of ATP production per unit volume

$$T\dot{S} = \int \mathrm{d}\mathbf{r}\left(\sigma_{\alpha\beta}v_{\alpha\beta} + h_\alpha\frac{D}{D_t}p_\alpha + r\Delta\mu\right) \tag{9.21}$$

in entropy production we thus have contribution from non-uniform flow, contribution from non-uniform rotation of the polartization, and contribution from the hydrolysis of ATP. The expression we found is in the form of pairs of interdependent forces and generalized currents. For example, we saw for isotropic liquid $\sigma_{\alpha\beta} = 2\eta v_{\alpha\beta}$. Near equilibrium exist conditions on the relations between the forces and currents, which are called Onsager Relations.

8.2 Onsager (Reciprocal) Relations

We'll assume quantities $\{x_i\}_{i=1}^N$ which describe the dynamics neat equilibirium, where in equilibirium $x_i = 0$ (for example $\theta - \theta_0$ in ordered homogenous states in 2D). We'll denote $h_i = -\frac{\partial S_T}{\partial x_i}$ and assume that the dynamics is described by a linear system:

$$\dot{x_i} = -\sum_{k=1}^{n}\gamma_{ik}h_k \tag{9.22}$$

where $\{\gamma_{ik}\}_{i,k}$ is a matrix of kinetic coefficients. We'll denote using σ_i the sign of the change in x_i under time reversal. For example, $\overline{x_i(-t)} = \sigma_i \overline{x_i(t)}$. Onsager showed that

$$\gamma_{ik} = \sigma_i \sigma_k \gamma_{ki} \ \ \forall i, k \tag{9.23}$$

which is called the Onsager Relations. For i, k with opposite behavior under time reversal $\gamma_{ik} = -\gamma_{ki}$ and the coupling does not contribute to entropy production. A coupling of this kind is called reactive. i, k with the same behavior under time reverasl do have a contribution to entropy production, and this coupling is called dissipative. Since total entropy can only grow, the symmetric part of γ_{ik} is a nonnegative matrix. The dissipative coefficients are connected by a correlation functions, as we saw in FDT for the diagonal coefficients γ_{ii}. Onsager Relations let us uniformly understand dynamic phenomenon which tie together different physical quantities, such as electric current from a temperature difference and currents of heat from a fall in voltage.

8.3 Construction of Constitutive Equations from Onsager Relations

We start with the entropy production rate

$$T\dot{S} = \int \mathrm{d}\mathbf{r} \left(\sigma_{\alpha\beta} \underline{v_{\alpha\beta}} + \underline{h_\alpha} \frac{D}{D_t} p_\alpha + \underline{r \Delta\mu} \right) \tag{9.24}$$

we'll call the underlined elements forces and describe them under time reversal and phenomenologically develop the 'currents' $r, \sigma_{\alpha\beta}, \frac{D}{Dt}P_\alpha$.

Force	Time Reversal effect	Associated Current
$\Delta\mu$	+	$r = r^r + r^d$
$v_{\alpha\beta}$	-	$\sigma_{\alpha\beta} = \sigma^r_{\alpha\beta} + \sigma^d_{\alpha\beta}$
h_α	+	$\frac{D}{Dt}P_\alpha = \left(\frac{D}{Dt}P_\alpha\right)^r + \left(\frac{D}{Dt}P_\alpha\right)^d$

Where the current elements superscipted with r are the reactive elements and those with d are the dissipative elements. We'll need to construct linear relations between scalars, vectors, and tensors. The vectors we have are $\mathbf{P}$ and ∇. $\mathbf{P}$ is more significant in the hydrodynamic limit. The tensors we have are $Q_{\alpha\beta}$, $\delta_{\alpha\beta}$, and $\partial_\alpha \partial_\beta$ and the such but these are less significant. We'll focus on the incompressible case, where P is a lagrange multiplier and we can focus only on $\sigma_{\alpha\beta}$ without its trace.

Dissipative Currents

Since the only force with negative time reversal is the velocity element, the dissipative part of its current is

$$\tilde{\sigma_{\alpha\beta}} = 2\eta v_{\alpha\beta} \tag{9.25}$$

the elements for h and μ both have positive time reversal symmetry so the dissipative current for those is

$$\left(\frac{D}{Dt}P_\alpha\right)^d = \frac{1}{\gamma_1} h_\alpha + \lambda P_\alpha \Delta\mu \tag{9.26}$$

the first coefficient is called the rotational viscousity. the other disspiative current is

$$r^d = \Lambda\Delta\mu + \lambda P_\alpha h_\alpha \tag{9.27}$$

where the first coefficient is called the hydrolysis efficiency. The equation for r will not appear in the hydrodynamic theory and we'll ignore it from now on. The final coupling λ is unimportant in ordered phase since $P^2 = 1$ and there is a lagrange multiplier in the dynamics in the direction P_α. In other words, the element parallel to P_α in $D_t P_\alpha$ won't appear in the equation of motion. This all means we can ignore equation 9.27 entirely and the 2nd element of equation 9.26.

Lecture 10.

Wed, January 14th, 2026

Reactive Currents

our equations we care about are

$$\sigma^r_{\alpha\beta} = -\zeta Q_{\alpha\beta}\Delta\mu + \frac{\nu}{2}\left(h_\alpha p_\beta + h_\beta P_\alpha - 2h_\gamma P_\gamma \frac{\delta_{\alpha\beta}}{d}\right) \tag{10.1}$$

$$\left(\frac{D}{Dt}P_\alpha\right)^r = -\nu P_\beta v_{\alpha\beta} \tag{10.2}$$

The first element of the first equation is called the active stress. It is proportional to $\Delta\mu$ which is the chemical potential difference which is what makes the system active. The right hand side of the 2nd equation is called the shear alignment. It describes how deformations in the flow rotate P_α. Note: in 2D, in shear flow with $\mathbf{P} = (\cos\theta, \sin\theta)$ we find a steady state with $\cos 2\theta = \frac{1}{\nu}$. There is a steady state like this only for $|\nu| > 1$. Otherwise, the particles will rotate indefinitely under shear flow (tumbling instability).

Notes

- The only active element is the active stress $\sigma^{ac}_{\alpha\beta} = -\zeta Q_{\alpha\beta}\Delta\mu$. For dry active matter we wrote $j^a_\alpha = -\Lambda\partial_\beta Q_{\alpha\beta}$. We can interpret this as a forces equation with friction on a surface: $\Gamma_{j\alpha} = \partial_\beta\sigma^{ac}_{\alpha\beta}$. We'll notice the 'mutual' element of ζ in σ is not part of the theory. In practice, we don't require Onsager relations for it.
- There is no guarantee that the system is near equilibirium and that Onsager relations hold. This is an uncontrolled approximation. We take consistent equations for a passive system and phenomenologically add active elements.

8.4 Development Of The Expression For Entropy Production

For a passive system with no flow we had

$$T\dot{S} = \frac{\mathrm{d}F}{\mathrm{d}t} = \int \mathrm{d}\mathbf{r}\left(\mu\partial_t\rho - h_\alpha\partial_t P_\alpha\right) \tag{10.3}$$

We'll want to reach the expression

$$T\dot{S} = \int \mathrm{d}\mathbf{r}\left(\sigma_{\alpha\beta}v_{\alpha\beta} + h_\alpha\frac{D}{Dt}P_\alpha\right) \tag{10.4}$$

To take into consideration the flow, we'll also look at the change in time of the kinetic energy of the subsystem using the convservation laws for mass and momentum. We have:

$$\frac{\partial}{\partial t}\left(\frac{1}{2}\rho v^2\right) = \frac{1}{2}\left[v_\alpha\partial_t\left(\rho v_\alpha\right) + v_\alpha\left(\partial_t\left(\rho v_\alpha\right) - v_\alpha\partial_t\rho\right)\right] \tag{10.5}$$

$$= v_\alpha\partial_t\left(\rho v_\alpha\right) - \frac{1}{2}v^2\partial_t\rho \tag{10.6}$$

so the change in time of the free energy plus kinetic energy is

$$\dot{F} = \int \mathrm{d}\mathbf{r}\left(\mu\partial_t\rho - h_\alpha\frac{D}{D_t}P_\alpha + h_\alpha\left(v_\beta\partial_\beta P_\alpha + \omega_{\alpha\beta}P_\beta\right) - \frac{1}{2}v^2\partial_t\rho + v_\alpha\partial_t\left(\rho v_\alpha\right)\right) \tag{10.7}$$

Using the conservation laws we find

$$T\dot{S} = \int \mathrm{d}\mathbf{r}\left[\underline{-\mu\partial_\alpha\left(\rho v_\alpha\right)} - h_\alpha\frac{D}{D_t}P_\alpha + h_\alpha\left(v_\beta\partial_\beta P_\alpha + \omega_{\alpha\beta}P_\beta\right) + \underline{\frac{1}{2}v^2\partial_\alpha\left(\rho v_\alpha\right)} + \underline{\underline{v_\alpha\partial_\beta\sigma^{tot}_{\alpha\beta}}}\right] = \tag{10.8}$$

where the element where mass conservation helped in is marked with an underline and the one for momentum conservation is double underlined. Using integration by parts

$$= \int \mathrm{d}\mathbf{r} \left[v_\alpha \left(\rho \partial_\alpha \mu + h_\beta \partial_\alpha P_\beta \right) - h_\alpha \frac{D}{D_t} P_\alpha + \omega_{\alpha\beta} h_\alpha P_\beta - \sigma^T_{\alpha\beta} \partial_\beta v_\alpha - \rho v_\alpha v_\beta \partial_\alpha v_\beta \right] \tag{10.9}$$

Inserting the Gibbs-Duhem relation

$$\mathrm{d}\Pi = \rho \, \mathrm{d}\mu + h_\beta \, \mathrm{d}P_\beta \implies \partial_\alpha \Pi = \rho \partial_\alpha \mu + h_\beta \partial_\alpha P_\beta \tag{10.10}$$

and breaking apart $\partial_\alpha v_\beta = v_{\alpha\beta} + \omega_{\alpha\beta}$ where v is the symmetric part and ω is the antisymmetric part, we find

$$T\dot{S} = \int \mathrm{d}\mathbf{r} \left[v_\alpha \partial_\alpha \Pi - h_\alpha \frac{D}{Dt} P_\alpha + \omega_{\alpha\beta} h_\alpha P_\beta - \sigma^{tot}_{\alpha\beta} \left(v_{\alpha\beta} - \omega_{\alpha\beta} \right) - \rho v_\alpha v_\beta \left(v_{\alpha\beta} + \omega_{\alpha\beta} \right) \right] \tag{10.11}$$

The trace of the multiplication of a symmetric and antisymmetric tensor is zero, so $\sigma^{tot}_{\alpha\beta} \omega_{\alpha\beta} = 0$ and $\rho v_\alpha v_\beta \omega_{\alpha\beta} = 0$. And breaking apart σ into a symmetric and antisymmetric part, we find

$$T\dot{S} = \int \mathrm{d}\mathbf{r} \left[- \left(\sigma^{T,S}_{\alpha\beta} + \rho v_\alpha v_\beta + \Pi \delta_{\alpha\beta} \right) v_{\alpha\beta} - h_\alpha \frac{D}{D_t} P_\alpha + \omega_{\alpha\beta} \left(\sigma^{T,AS}_{\alpha\beta} + \left(h_\alpha P_\beta \right)^{AS} \right) \right] \tag{10.12}$$

We know that in the rotation of a solid body, $\dot{F} = 0$ so the last element should be zero. We'll conclude that there is an antisymmetric part of the stress tensor

$$\sigma^{T,AS}_{\alpha\beta} = \frac{1}{2} \left(P_\alpha h_\beta - P_\beta h_\alpha \right) \tag{10.13}$$

8.5 Conclusion

The final expression for entropy production is

$$T\dot{S} = \int \mathrm{d}\mathbf{r} \left(\sigma_{\alpha\beta} v_{\alpha\beta} + h_\alpha \frac{D}{Dt} P_\alpha + r \Delta\mu \right) \tag{10.14}$$

and our equations are:

$$\partial_\beta \sigma^{tot}_{\alpha\beta} = 0 \tag{10.15}$$

$$\left(\partial_t + v_\beta \partial_\beta \right) P_\alpha = \frac{h_\alpha}{\gamma_1} - \left(\omega_{\alpha\beta} + \nu v_{\alpha\beta} \right) P_\beta \tag{10.16}$$

where

$$h_\alpha = h_\| P_\alpha + K \partial_{\beta\beta} P_\alpha \tag{10.17}$$

and

$$\sigma^{tot}_{\alpha\beta} = -\zeta \Delta\mu Q_{\alpha\beta} + 2\eta v_{\alpha\beta} - \Pi \delta_{\alpha\beta} - \cancel{\rho v_\alpha v_\beta} + \frac{\nu + 1}{2} P_\alpha h_\beta + \frac{\nu - 1}{2} h_\alpha P_\beta \tag{10.18}$$

$$\left(\partial_t + v_\beta \partial_\beta \right) P_\alpha = \frac{h_\|}{\gamma_1} P_\alpha + \frac{K}{\gamma_1} \partial_{\beta\beta} P_\alpha - \left(\omega_{\alpha\beta} + \nu v_{\alpha\beta} \right) P_\beta \tag{10.19}$$

The first element in the first equation is the active stress, the second is the viscous stress, the 3rd is the isotropic pressure (which is a lagrange multiplier), the 4th is the convective momentum flux which we consider negligible, and the last two are stress from rotation and alignment of the polar order parameter. In the 2nd equation, the first element is a lagrange multiplier to enforce $|\mathbf{P}| = 1$, the 2nd is diffusion of the order parameter, and the last is rotation and alignment as a result of the flow.

8.6 Spontaneous Flow in Active Nematic Matter

The equations have a steady state with nematic order in an arbitrary direction $\hat{x}$ and $\mathbf{v} = 0$. i.e the system doesn't flow. In certain conditions this state is unstable and Spontaneous flow appears. How does that happen? When there's a gradient in the nematic order, the active stress creates flow. This flow rotates the nematic field and allows it to remain inhomogeneous.

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